

MINI - AIR COMPRESSOR

ABSTRACT:

Our project based on mini air compressor have relatively simple design and can be sized all the way from the extremely powerful to ones that can just drive small tools. They have few parts and hence last for a longtime with little maintenance. The factors make them drivers of choice from heavy equipment on manufacturing shop floors, to cleaning areas in gas stations and now, inside homes and small garages as parts of handy gadgets

INTRODUCTION:

Compressor is a device used to increase the low pressure to high pressure by using some external energy. For filling the air tube of vehicles, in automobile service station to clean vehicles, for spray painting in paint industries, in vehicles to operate air brakes, for cleaning workshop machines, for supercharging of an ic engines, for operation of pneumatic tools I.e. rock drills, tubes ,vibrations etc.

An air compressor works by transforming electrical energy in to kinetic energy in the form of air shaft is held in a highly compressed space before sudden release. The energy released when the compressed air is let out can be used for inflation cleaning under pressure, turning, generating torque, driving or other similar movement by force. COMPONENTS REQUIRED: *12V Dc Gear Motor (Geared Motor) 300 Rpm

*Syringe *wood

*valve pins *cycle forks

*wire tube *switch

*charger (charging macforks) *screws pins



12V-DC MOTOR



WIRE TUBE



ELETRICAL CHARGER



ELETRICAL SWITCH



CYCLE SPOKES



MINI AIR COMPRESSOR

METHODOLOGY

Market research Decision on capacity and pressure of the compressor Power Calculation calculations of compressor's parts Choose motor unit Assembly Manufacturing issues Lubrication Safety, Maintenance

FUNCTION OF COMPRESSOR

The function of a compressor is to take a definite quantity of fluid (usually a gas, often Air) and deliver it at a required pressure. Reciprocating type – low mass flow rate (kg/s) and high-pressure ratio. Rotary type – high mass rate but low pressure ratio” PMV

$$m = P/V$$

ADVANTAGES & DISADVANTAGES

ADVANTAGES:

- *They tend to be less expensive
- *Upgrading permanent mount compressors with air filters, water filters, and in-line lubrication is Easier.
- *There is a much broader selection of electrical power options for mounted units
- *Generally, they are more powerful.
- *Generally, they have greater air capacity.

DISADVANTAGES:

- *Not portable
- *More air hose must be run when distance is in issue and that means some additional cost

APPLICATIONS:

- *Painting vehicles in an auto body shop
- *Sanding in an auto body shop or in woodworking
- *Using pneumatic nail guns for roofing
- *Providing dental and medical services
- *Using pneumatic drills and hammers on construction sites
- *Powering various air tools in an automotive repair shop
- *Using an air blowgun to clean machinery
- *Sandblasting in a machine shop and manufacturing facilities

RESULT AND DISCUSSION

Air compressors have a wide range of uses at home and in industrial settings. Although common only in manufacturing plants before, they are now used in home basements and garages as go-to device. They act as a power source for electric tools

SUMMARY:

The summary of our project is to create mini air compressor which is developed by a Compressed air by using the 12V- Dc motor, syringes,chargers etc. An air compressor works by Transforming electrical energy in to kinetic energy in the form of air that is held in a highly Compressed space before sudden release. It's easy to maintain, lower in cost, higher production Rate. ACKNOWLEDGEMENTS:

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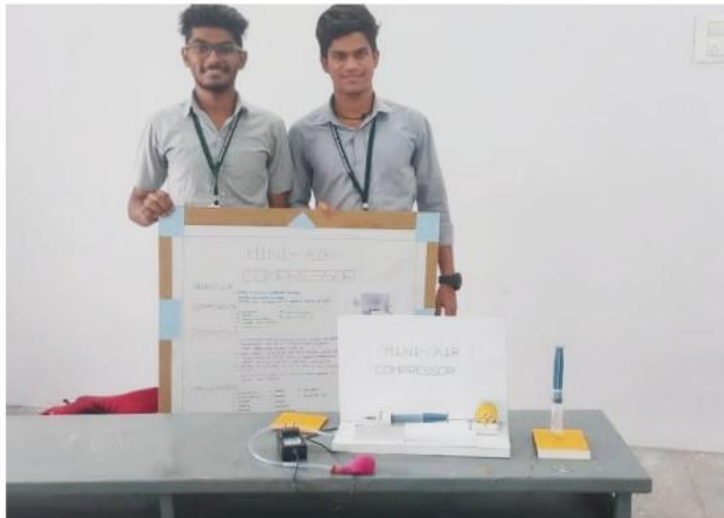
We thank Prof. P. Suresh Varma,

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